



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

B.Tech: IV – SEMESTER

COURSE CODE	COURSE NAME	TH.	TU.	PR.	CREDITS	EVALUATION		
401BCE24	DATA STRUCTURE	3	-	-	3	ESE	CIE	TOTAL
						60	40	100

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: ❖ To import the Notations and basic concepts of Data Structure and Algorithm. ❖ To emphasize the application of Data Structure in developing and implementing efficient programs and algorithms using searching and sorting. ❖ To understand the basic concepts Stack & Queue.	After completing this course, the student will be able to: CO1: Understand the basic concepts of data structures, their classification, and their role in solving computational problems. CO2: Implement various searching algorithms like linear search and binary Ssearch, and compare their performance same as sorting. CO3: Understand the concepts, operations, and applications of stack and queue data structures. CO4: Implement different types of linked lists such as singly, doubly, circular singly, and circular doubly linked lists. CO5: Understand the basic terminology, structure, and properties of tree data structures. CO6: Implement graph structures for both directed and undirected graphs using appropriate data structures.

Unit I:

(07 Hours)

INTRODUCTION TO DATA STRUCTURE

General concepts of data structure: properties and Operations, Time and space analysis of algorithms, Big-Oh Notation, Theta Notation, and Omega Notations, Average, best and worst case analysis, Abstract data structure.

Hashing: Hash tables, hash functions, hashing techniques, and Collision resolution techniques.

Unit II:

(07 Hours)

SEARCHING, SORTING AND ALGORITHM'S

Searching, Sorting and Algorithm Techniques: Selection Sort, Insertion Sort, Merge Sort, Shell Sort, Radix Sort, Linear and Binary Search.

Unit III:

(07 Hours)

STACK & QUEUE

Representation of Stack & queue using array: Application of Stacks, Conversion from Infix to Postfix and Prefix expressions, Evaluation of Postfix, Circular Queues, Linear Queue and Priority Queues.

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Unit IV:

(08 Hours)

LINKED LIST

Definition and representation in memory, Implementation of Linked List, Types of linked list: Singly linked list, Circular Singly linked list, Doubly linked list and Circular Doubly linked list and Representation of Ordered list using array and its operation, Implementation of linked list using Static and Dynamic Memory Allocations, Operations of linked list (insertion, deletion, searching, traversing.) Application of linked list such as polynomial expression, comparisons of linked lists.

Unit V:

(07 Hours)

TREES

Definition and basic terminology, Representation of tree. Basic operations of binary trees and binary search Trees (traversals of trees, insertion and deletion of elements). Binary search trees the concept of balancing. Threaded Binary Trees, AVL Trees, Heap trees, B-Trees.

Unit VI:

(05 Hours)

GRAPHS

Representation of Graph, Matrix Representation of Graph, List Representation of Graph, Directed Graphs, graph traversal (BFS and DFS) with complexity analysis, shortest path, Spanning trees.

Text Book:

- 1) **Data Structures**, Edition 2023-24, Publisher: & Distributor Sia.
- 2) **Data Structures And Algorithms**, 2024, Huda Noordean, Publisher: Notion Press.
- 3) **Problem Solving In Data Structures & Algorithms Using C**, 2022, Publisher: Hemant Jain.

Reference Books:

- 1) **Data Structures** Third Edition, 2023, Reema Thareja Publisher: Oxford.
- 2) **Data Structure Algorithms and Analysis**, 2021 Dr. Ashwin Mehta Publisher: Spd.
- 3) **Data Structure Through C**, 2023 Yashvant Kanetkar Publisher: Computer And It Books.

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B.Tech: IV– SEMESTER

COURSE CODE	COURSE NAME	TH.	TU	PR	CREDITS	EVALUATION		
402BCE24	DATA STRUCTURE-LAB	-	-	02	01	ESE	CIE	TOTAL
						25	25	50

COURSE OBJECTIVE	COURSE OUTCOME
The objectives of this course is to impart knowledge of : ❖ To Import the Notations and basic concepts of Data Structure and Algorithm. ❖ To emphasize the application of Data Structure in developing and implementing efficient programs and algorithms using searching and sorting. ❖ To understand the basic concepts Stack & Queue.	After completing this course, the student will be able to: CO1: Understand the basic concepts of data structures. their classification, and their role in solving computational problems. CO2: Implement various searching algorithms like linear search and binary Ssearch, and compare their performance same as sorting. CO3: Understand the concepts, operations, and applications of stack and queue data structures. CO4: Implement different types of linked lists such as singly, doubly, circular singly, and circular doubly linked lists. CO5: Understand the basic terminology, structure, and properties of tree data structures. CO6: Implement graph structures for both directed and undirected graphs using appropriate data structures.

Write a concept of Data Structure

- 1) To study the concept of Data Structure and Asymptotic Notations (best, worst, and average cases)

Programs on Searching and sorting

- 2) To implement a program for sorting methods with Bubble Sort or Insertion sort
- 3) To implement a program for recursive Linear Search and Binary Search methods.

Programs on Stack and Queue

- 4) To implement a Program for Stack (its operations) using Array.
- 5) To implement a Program for Queue (its operations) using Array
- 6) To implement a Program for Linear Queue. Circular Queues, and Priority Queues

Programs on Linked List

- 7) To implement a Program for the following operations on singly linked list
i) Creation ii) Insertion iii) Deletion iv) Traversal.
- 8) To implement a Program for linked list using Static and dynamic memory allocations.
- 9) To implement a Program for circular queue as a Linked List.

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Tree

- 9) To implement a Program for Binary Search Tree.
- 10) To implement AVL Tree.

Graph

- 11) To implement a Program for graph using adjacency list
- 12) To study the concept of Graph representation
- 13) To implement a Program for BFS and DFS traversal.
- 14) To implement a Program for Dijkstra's algorithm for both directed and undirected graphs.

Text Books:

- 1) Data Structures & Algorithms using C. 5th Edition, 2024, R.S. Salaria, Publisher: Khanna Publishing House.
- 2) Introduction to Algorithms, 4th Edition, Cormen. Leiserson, Rivest, Stein, Publisher: MIT Press
- 3) Problem Solving In Data Structures & Algorithms Using C. 2022. Publisher: Hemant Jain.

Reference Books:

- 4) Data Structures using C Third Edition, 2020, Reema Thareja Publisher: Oxford University Press.
- 5) Data Structures and Algorithm Analysis in C, 2nd 3rd Edition, 2020 Mark Allen Weiss Publisher: Pearson Education.
- 6) Data Structures and Algorithms Made Easy, 2nd/3rd Edition (C Version), 2020-2021 editions available Narasimha Karumanchi Publisher: Career Monk Publications.
- 7) Data Structures using C and C++, Latest reprints (Original 2nd Edition), Reprints available till 2023 Tannenbaum. Langsam, and Augenstein Publisher: Pearson Education.

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B.Tech: IV – SEMESTER

COURSE CODE	COURSE NAME	TH.	TU.	PR.	CREDITS	EVALUATION		
403BCE24	OPERATING SYSTEM	02	-	-	02	ESE	CIE	TOTAL
						30	20	50

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] Understand the fundamental concepts, functions, and evolution of operating systems. [2] Understand process management, scheduling algorithms, and inter-process communication. [3] To explore memory management techniques, including paging, segmentation, and virtual memory. [4] To analyze file system structures, storage management, and directory organization.	After completing this course, the student will be able to: CO1. Explain the fundamental concepts, functions, and evolution of operating systems. CO2. Demonstrate an understanding of process management, scheduling algorithms, and inter-process communication. CO3. Analyze memory management techniques, including paging, segmentation, and virtual memory, for efficient memory utilization. CO4. Describe file system structures, storage management techniques, and directory organization methods. CO5. Identify and apply deadlock detection, prevention, and avoidance strategies along with synchronization mechanisms. CO6. Evaluate advanced operating system concepts such as distributed computing, virtualization, cloud computing, and security mechanisms.

Unit I:

(7 Hours)

INTRODUCTION TO OPERATING SYSTEMS

Definition and functions of an Operating System, Evolution of Operating Systems, Types of Operating Systems (Batch, Multiprogramming, Time-Sharing, Distributed, Real-Time, Mobile OS), System Calls and their types Operating System Services

Unit II:

(8 Hours)

PROCESS MANAGEMENT

Process Concept and Process States, Process Control Block (PCB), Process Scheduling (Long-term, Short-term, Medium-term), Scheduling Algorithms, Threads and Multithreading, Process Synchronization, Inter-Process Communication (IPC)

Unit III:

(6 Hours)

MEMORY MANAGEMENT

Memory Hierarchy and Management Strategies, Contiguous Memory Allocation, Non-Contiguous Allocation, Virtual Memory Concepts, Demand Paging, Page Replacement Algorithms (FIFO, LRU, Optimal)

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DIPLOMA PROGRAM
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Unit IV:

(6 Hours)

FILE SYSTEM & STORAGE MANAGEMENT

File System Concepts and Operations, File Access Methods (Sequential, Indexed, Direct), File Allocation Techniques (Contiguous, Linked, Indexed), Directory Structures (Single-level, Two-level, Tree-structured, Acyclic, Graph)

Unit V:

(7 Hours)

DEADLOCKS & SYNCHRONIZATION

Deadlock Characterization, (Mutual Exclusion, Hold and Wait, No Preemption, Circular Wait), Deadlock Prevention, Avoidance (Banker's Algorithm), Detection, and Recovery, Semaphores and Monitors

Unit VI:

(7 Hours)

DISTRIBUTED COMPUTING AND NETWORKED OPERATING SYSTEMS

Distributed Operating Systems, Virtualization and Cloud Computing, Security, Privacy, and Protection in Modern Operating Systems.

Text Books:

- 1) **Operating System Concepts**. Abraham Silberschatz, 30 July 2019, Wiley
- 2) **Operating System Concepts** by Abraham Silberschatz, 8th edition, Wiley

Reference Books:

- 1) **Operating Systems Internals And Design** WILLIAM STALLINGS 9th Edition, Pearson India
- 2) **Modern Operating Systems** – Andrew S. Tanenbaum, 5th Edition, Pearson Education.
- 3) **Operating System** - DEITEL , 3rd Edition, Pearson India.

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B.Tech : IV – SEMESTER

COURSE CODE	COURSE NAME	TH.	TU	PR	CREDITS	EVALUATION		
404BCE24	OPERATING SYSTEM - LAB	-	-	02	01	ESE	CIE	TOTAL
						25	25	50

COURSE OBJECTIVE	COURSE OUTCOME
The objectives of this course is to impart knowledge of 1. Understand the fundamental concepts, functions, and evolution of operating systems. 2. Understand process management, scheduling algorithms, and inter-process communication. 3. To explore memory management techniques, including paging, segmentation, and virtual memory. 4. To analyze file system structures, storage management, and directory organization. 5. To study deadlock detection, prevention, avoidance strategies, and synchronization mechanisms. 6. To explore advanced operating system concepts, including distributed computing, virtualization, cloud computing, and security mechanisms.	After completing this course, the student will be able to: CO1. Explain the fundamental concepts, functions, and evolution of operating systems. CO2. Demonstrate an understanding of process management, scheduling algorithms, and inter-process communication. CO3. Analyze memory management techniques, including paging, segmentation, and virtual memory, for efficient memory utilization. CO4. Describe file system structures, storage management techniques, and directory organization methods. CO5. Identify and apply deadlock detection, prevention, and avoidance strategies along with synchronization mechanisms. CO6. Evaluate advanced operating system concepts such as distributed computing, virtualization, cloud computing, and security mechanisms.

PROCESS MANAGEMENT

1. Write a c program to simulate the CPU scheduling algorithm First Come First Serve (FCFS).
2. Write a program to stimulate the CPU scheduling algorithm Shortest job first (Non- Preemption)
3. To simulate the CPU scheduling algorithm round-robin.
4. To write a c program to simulate the CPU scheduling priority algorithm
5. Write a C program to simulate producer-consumer problem using semaphores.
6. To Write a C program to simulate the concept of Dining-Philosophers problem

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MEMORY MANAGEMENT

7. To implement and simulate the MFT algorithm
8. To write a program to simulate the MVT algorithm
9. To Write a C program to simulate the following contiguous memory allocation techniques a) Worst-fit b) Best-fit c) First-fit

PAGE REPLACEMENT ALGORITHMS

10. To implement FIFO page replacement technique. a) FIFO b) LRU c) OPTIMAL

FILE SYSTEM & STORAGE MANAGEMENT

11. Program to simulate Single level directory file organization technique
12. Program to simulate two level file organization technique

DEADLOCKS & SYNCHRONIZATION

13. To Simulate banker's algorithm for Dead Lock Avoidance (Banker's Algorithm)

Text Books:

1. **Operating System Concepts**. Abraham Silberschatz, 30 July 2019, Wiley
2. **Operating System Concepts** by Abraham Silberschatz, 8th edition, Wiley

Reference Books:

- 4) **Operating Systems Internals and Design** WILLIAM STALLINGS 9th Edition, Pearson India
- 5) **Modern Operating Systems** – Andrew S. Tanenbaum, 5th Edition, Pearson Education.
- 6) **Operating System** - DEITEL, 3rd Edition, Pearson India.

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B.Tech: IV - SEMESTER

COURSE CODE	COURSE NAME	TH.	TU.	PR.	CREDITS	EVALUATION		
405BCE24	THEORY OF COMPUTATION	02	-	-	02	ESE	CIE	TOTAL
						30	20	50

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: ❖ To introduce the fundamental concepts of formal languages and automata theory, including finite automata, regular languages, and the Chomsky hierarchy. ❖ To familiarize students with context-free languages, normal forms for context-free grammars, and the Post Correspondence Problem..	After completing this course, the student will be able to: CO1: Demonstrate the fundamental concepts of formal languages, finite automata, and regular expressions. CO2: Design deterministic and non-deterministic finite automata, and perform minimization. CO3: Apply concepts of context-free languages to simplify grammars using reductions and normal forms. CO4: Explain the theory of Turing machines, recursive functions, decidability, and the Post Correspondence Problem.

UNIT I

(07 Hours)

Concept of language, Strings, formal grammars, Chomsky hierarchy. Finite Automaton, regular languages, deterministic & non deterministic finite automata, minimization of automata.

UNIT II

(08 Hours)

Finite automata with E-closures, equivalence, Regular expression, Arden theorem, closure & decision properties for regular sets, Equivalence of RE and FA, Interconversion, Regular grammar, right linear, left linear, Interconversion of grammar.

UNIT III

(07 Hours)

Context free grammar, Context free languages, Derivation tree, Parse tree, reduction of CFGS, Normal forms for CFG, Chomsky normal form, Greibach normal form, Push down Automata (PDA), definition, equivalence of CFL & PDA

UNIT IV

(08 Hours)

Introduction to Turing Machine, Types of TM, Undecidability, Decidability Post correspondence Problem (PCP), Modified Post correspondence Problem (MPCP), Introduction to primitive recursive function (PRF), Ackerman functions.

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Text Book:

- 1) Introduction Of Automata Theory, Languages and computation- Hopcroft, Motwani & Ulman
- 2) Introduction to formal languages and automata – Peter Linz.
- 3) Introduction to Theory of Computation – Michael Sipser.

Reference Books:

- 1) Introduction to Languages and the Theory of Computation by John C. Martin
- 2) Foundations of Computation by Carol Critchlow and David J. Eck
- 3) Theory of Computation by Dr. O.G. Kakade

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DIPLOMA PROGRAM
CE & ME

B.Tech: IV SEMESTER

COURSE CODE	COURSE NAME	TH	TU	PR	CREDITS	EVALUATION		
406BCE24-A	INTRODUCTION TO DATABASE MANAGEMENT SYSTEM (Open Elective –II)	2	-	-	2	ESE	CIE	TOTAL
						30	20	50

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] Introduce the fundamental concepts of Database Management Systems (DBMS): [2] Understand Data Models and their Importance. [3] To apply the fundamentals of SQL for database management.	After completing this course, the student will be able to: CO1: Understand the basic of DBMS to analyze the information and also design the ER model. CO2: Explain the importance of different types of data models and Key Concept. CO3: Demonstrate SQL queries to perform database Operations. CO4: Explore the concept of functional dependency, various normalization forms and database transactions.

UNIT I

(7 Hours)

INTRODUCTION TO DBMS

Purpose of database system, file system Vs DBMS, characteristics of database system, Database system application, view of data Three-schema architecture of a database, DBMS Architecture-Variou components of a DBMS, database design using ER Model: entity, attribute, relationship, ER Diagram.

UNIT II

(7 Hours)

DATA MODELS

Importance of data model, Types of Data Model, Relational Data Model: Concept of relations, domain, tuple, Schema-instance distinction, constraints and Keys.

UNIT III

(7 Hours)

CONCEPT OF SQL

Basic of SQL, DDL, DML, DCL, TCL, Aggregate function, built -in-function-Numeric, date, string, Different types of clauses: Group By, order By, Having, join operation and its types, nested sub queries.

UNIT IV

(7 Hours)

NORMALIZATION AND TRANSACTION MANAGEMENT

Concepts of Functional dependency, Data integrity rules. Normalization: concept of normalization, 1NF, 2NF, 3NF, BCNF. Transaction concepts, ACID properties, State of transactions, concurrency control.

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Text Books:

- 1) **Database System Concepts** by Avi Silberschatz, Henry F. Korth. S. Sudarshan, Tata McGraw Hill, Fifth Edition.
- 2) **Fundamentals of Database Management Systems**, 7th Edition by Ramez Elmasri, Shamkant B. Navathe 2020.
- 3) **Fundamentals of Database Management Systems, 3rd Edition** by Mark L. Gillenson 2023.
- 4) **Database Management Systems**, 3rd Edition by Raghu Ramakrishnan and Johannes Gehrke 2003.

Reference Books:

- 1) **Introduction to database management system**, 4th Edition by kahate Pearson Education.
- 2) **Database Systems-Design, Implementation and Management (7/e)** by Peter Rob and Carlos Coronel, 4th Edition, Cengage Learning.

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B.Tech: IV SEMESTER

COURSE CODE	COURSE NAME	TH	TU	PR	CREDITS	EVALUATION		
406BCE24-B	COMPUTER NETWORKS (Open Elective –II)	3	-	-	03	ESE	CIE	TOTAL
						60	40	100

COURSE OBJECTIVE	COURSE OUTCOME
The students will try to learn ❖ Understand the Fundamentals of Computer Networks ❖ Understand the concept of layered models and protocol hierarchies	At the end of the course students should be able to: CO1: Identify different types of networks (LAN, MAN, WAN, PAN, CAN) and their applications. CO2: Analyze protocol hierarchies and their role in network communication. CO3: Understand IPv4 and IPv6 addressing and analyze their functionalities. CO4: Compare and contrast connection-oriented (TCP) and connectionless (UDP) transport protocols.

Unit 1:

(6 Hours)

INTRODUCTION TO COMPUTER NETWORKS: Definition, Data Representation, Types of Data flow, Need of Computer Networks, advantages and disadvantages, Goals and Application of Computer Network, Network Hardware Components, Topology with its types, Types of networks: (LAN, MAN, WAN, PAN, CAN), Transmission Impairment, OSI Reference Model, TCP/IP model.

Unit II:

(9 Hours)

PHYSICAL & DATA LINK LAYER

Guided media & unguided media, twisted pair coaxial Cable, fiber optics. Wireless transmission, microwave transmission. Introduction: Design Issues with data link layer, Flow Control & Error Control Protocols: Stop-and-Wait ARQ, Go-Back-N ARQ, Selective Repeat ARQ, Sliding Window Protocol, virtual LAN.

Unit III:

(7 Hours)

NETWORK LAYER

Services provided to the Transport Layer, Network Layer Design Issues, Switching Techniques, Internet Protocol (IP): IPv4, IPv6, ARP, RARP, DHCP, Routing Protocols BGP, OSPF, ICMP, IGMP, Traffic Shaping: Leaky Bucket, Token Bucket.

Unit IV:

(8 Hours)

TRANSPORT & APPLICATION LAYER

Functions of the Transport Layer, Connection-oriented vs Connectionless services, Open-loop and Closed-loop Congestion Control, Quality of Service (QoS) parameters, Overview of TCP, UDP, HTTP, FTP, DNS

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Text Books

- 1) An Engineering Approach to Computer Networking By Keshav S Publisher Pearson Education 2nd Edition
- 2) Computer Networks and Internet By Comer D Publisher Pearson Education 2nd Edition
- 3) Computer Networks" by Andrew S. Tanenbaum
- 4) Data Communications and Networking" by Behrouz A. Forouzan

Reference Books

- 1) Data Communications and Networking By Forouzan TataMcGraw-Hill Publications, 4 th Edition.
- 2) Computer Networking -A Top-Down Approach By James F. Kurose Pearson Publication, 7th Edition.

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DIPLOMA PROGRAM
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B.Tech: IV - SEMESTER

COURSE CODE	COURSE NAME	TH.	TU.	PR.	CREDITS	EVALUATION		
406BCE24-C	OPERATING SYSTEM (Open Elective-II)	02	-	-	02	ESE	CIE	TOTAL
						30	20	50

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: ❖ Understand the fundamental concepts, purpose, and functions of operating systems. ❖ Understanding memory management, file systems, and disk scheduling for efficient OS resource management.	After completing this course, the student will be able to: CO1: To build a foundation in OS principles for designing efficient and reliable system architectures CO2: Develop a strong understanding of memory management, file systems, and system calls. CO3: Analyze scheduling, process thread management, and concurrency control mechanisms for efficient program execution and disk scheduling techniques to effectively manage OS resources. CO4: Identify and explain the conditions leading to deadlock in operating systems & Memory Management

Unit I

(7 Hours)

Introduction to Operating Systems

Operating System Overview: Introduction, Types of operating systems (Simple Batch, Multi programmed, Time-shared, Personal Computer, Parallel, Distributed Systems, Real-Time Systems), Operating System Service, Operating System Structure.

Unit II

(8 Hours)

File System Operations & Deadlocks

File System Interface and Operations: Directory structure, access methods, protection mechanisms, file system organization, allocation methods, and free-space management

System Calls for File Operations: Usage of system calls like open, create, read, write, close, seek, stat for interacting with files.

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DIPLOMA PROGRAM
CE & ME

Unit III

(8 Hours)

Process Management & CPU Scheduling

Process Concepts & Operations: Process creation, scheduling, process states, and transitions

Process Scheduling: Criteria for scheduling, scheduling algorithms (Round Robin, FCFS, SJF), multiple processor scheduling

Unit IV

(7 Hours)

Memory Management & Deadlocks

Memory Management: Logical vs. physical address space, memory allocation strategies (contiguous, paging, segmentation), swapping, and segmentation with paging.

Deadlocks: Characterization, methods for handling deadlocks (prevention, avoidance, detection, recovery)

Text Book:

- 1) **Modern Operating Systems**, | By Andrew Tanenbaum, Herbert Bos ,Global 5th Edition, Kindle Edition, Publications-Pearson 5th edition (28 March 2023)
- 2) **Operating Systems Design And Implementat** | By Andrew S. Tanenbaum | 3rd Edition | Pearson Publication (English Medium)(2022)
- 3) **MODERN OPERATING SYSTEMS** | By Prof. Shraddha N. Zanjat ,Prof. V. K. Barbudhe, Dr. Bhavana S. Karmore Universe-Win Publication House and Research Services Pvt. Ltd.(2022)

Reference Books:

- 1) **Operating System Concepts** | By Abraham Silberschatz (Author), Peter B. Galvin (Author), Greg Gagne (Author), Publication Silberschatz's (2020)
- 2) **Operating Systems: A Concept-Based Approach** | By Dhananjay Dhamdhare (Author), Publication Mc Graw Hill(2015)

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DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

B.Tech: IV SEMESTER

COURSE CODE	COURSE NAME	TH	TU	PR	CREDITS	EVALUATION		
407BCE24	ENTREPRENEURSHIP IN COMPUTER ENGINEERING	2	-	-	2	ESE	CIE	TOTAL
						30	20	50

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] To introduce entrepreneurship fundamentals and tech startup opportunities in Computer engineering. [2] To understand product development, prototyping, and business models for tech startups. [3] To explore financial planning, marketing strategies, and legal aspects of startup growth.	After completing this course, the student will be able to: CO1: Understand entrepreneurship fundamentals and identify tech startup opportunities. CO2: Apply agile development and MVP techniques in product prototyping. CO3: Analyze business models, revenue streams, and financial planning strategies. CO4: Implement digital marketing, sales strategies, and legal compliance in tech startups.

UNIT -I

(7 Hours)

INTRODUCTION TO ENTREPRENEURSHIP & COMPUTER ENGINEERING

Entrepreneurship Basics: Definition & Meaning of Entrepreneurship, Importance of Entrepreneurship in Economic Growth, Key Characteristics of Entrepreneurs (Risk-taking, Innovation, Leadership, Decision-making)

Computer Engineering Industry Overview: Role of Computer Engineering in Entrepreneurship, Overview of Software Development, AI & ML, Cyber security, IoT

Tech Startup Opportunities: Emerging Fields: AI, Data Science, Block chain, Startup Ideas in Computer Engineering

Innovation & Creativity in Tech Startups: Ideation Techniques (Brainstorming, SCAMPER, Mind Mapping), Design Thinking Process (Empathize, Define, Ideate, Prototype, Test)

Startup Ecosystem: Role of Incubators & Accelerators, Government Schemes (Startup India, Digital India, MSME Support)

UNIT -II

(8 Hours)

PRODUCT DEVELOPMENT & PROTOTYPING

Opportunity Identification: Importance of Market Research, Competitive Analysis in Tech Startups

Product Development Process: Agile Development & its Stages, Minimum Viable Product (MVP) – Concept & Benefits

Prototyping & Deployment: Stages of Prototyping in Tech Startups, Testing Methods & Launch Strategies

Technology Stack for Startups: Cloud Computing (AWS, Google Cloud, Azure), DevOps Tools & APIs for Automation

Security & Compliance: Cyber security Best Practices for Startups, Data Protection & Privacy Laws

Intellectual Property (IPR): Copyrights & Patents in Software Development

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DIPLOMA PROGRAM
CE & ME

UNIT- III

(7 Hours)

BUSINESS MODEL & FINANCIAL PLANNING

Business Models in Tech Startups: Software-as-a-Service (SaaS), Platform-as-a-Service (PaaS), Infrastructure-as-a-Service (IaaS)

Revenue Streams: Subscription Model, Pay-Per-Use Model

Financial Planning: Basics of Budgeting for Startups, Break-Even Analysis (Understanding Cost vs. Revenue)

Funding Sources for Startups: Angel Investors & Venture Capitalists, Government Grants & Startup India Funding

Legal & Ethical Aspects: IT Act & Data Privacy Laws

Case Study: Example: Flipchart's Funding Journey, Ola's Business Model

UNIT -IV

(8 Hours)

MARKETING, SALES & SCALING TECH VENTURES

Marketing Strategies for Tech Startups: Introduction to Digital Marketing, SEO Strategies & Website Optimization

Sales Channels for Startups: B2B (Business-to-Business) & B2C (Business-to-Consumer) Models

Customer Acquisition & Retention: CRM (Customer Relationship Management) & User Engagement Strategies

Scaling a Startup: Strategic Partnerships & Collaborations, Global Expansion Strategies

Challenges & Risks in Tech Startups: Market Competition & Survival Strategies, Cyber security Threats & Data Breaches

Case Study: Example: Zomato's Global Expansion, Amazon's Scaling Strategy

Text Books

- 1) "Entrepreneurship" – By Rajeev Roy (Covers fundamentals of entrepreneurship, startup ecosystems, and innovation techniques.)
- 2) "Technology Ventures: From Idea to Enterprise" – By Thomas Byers, Richard Dorf, Andrew Nelson (Ideal for CE students, focuses on product development, scaling, and tech entrepreneurship.)
- 3) "The Startup Owner's Manual" – By Steve Blank & Bob Dorf (Practical guide for startup development and business models.)
- 4) "Lean Startup" – By Eric Ries (Best for MVP development, Agile methodology, and scaling strategies.)
- 5) "Data Science for Business" – By Foster Provost & Tom Fawcett (Useful for CE students exploring AI, ML, and Data Science startups.)

Reference Books

- 1) "Zero to One" – By Peter Thiel (Explores innovation, competitive advantage, and successful startups.)
- 2) "The Art of the Start 2.0" – By Guy Kawasaki (Focuses on launching and funding tech startups.)
- 3) "Business Model Generation" – By Alexander Osterwalder (Detailed insights on business models & revenue strategies.)
- 4) Startup India: The inside Story" – By N. K. Gupta & Shivani Gupta (Covers real-life Indian startup case studies and government policies like Startup India, Digital India.)
- 5) "Stay Hungry Stay Foolish" – By Rashmi Bansal (Success stories of 25 IIM-A entrepreneurs, great for motivation and Indian startup insights.)

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DIPLOMA PROGRAM
CE & ME

B.Tech: IV SEMESTER

COURSE CODE	COURSE NAME	TH	TU	PR	CREDITS	EVALUATION		
408BSH24	ENVIRONMENTAL SCIENCE	2	-	-	2	ESE	CIE	TOTAL
						30	20	50

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] To impart knowledge in the domain of Environmental issues due to industrial growth. [2] To bring out Impact of Technologies on Environment. [3] To inculcate knowledge about assessing the environmental issues and implementation of controlling methods on diverse effect of industrialization.	After completing this course, the student will be able to: CO1: Analyze the impacts of Industrial pollution and its control. CO2: Illustrate various types of pollutants and waste management. CO3: Analyze the impacts of air, water pollution and its control. CO4: Adopt sustainability as a practice in life, society and industry and know their roles and ethics, as citizens, and consumers in a complex, interconnected world.

UNIT -I

(7 Hours)

Environmental impacts of Energy Technologies

Industrial pollution due to non-renewable energy sources: General Introduction of Industrial pollution and its types. Principle, processes, source of pollution.

Environmental impact and its control with reference to specific industries; like Nitrogen containing fertilizers-ammonia synthesis, Cement manufacturing Industry; Sulfuric acid manufacturing industry and petroleum Industry

UNIT -II

(7 Hours)

Soil pollution and its control

Soil pollution: Soil around us, soil water characteristics, soil pollution

Causes effect and control: Noise pollution, nuclear and radiation hazards, marine pollution (Oil spills and ocean acidification).

Solid waste management: Composting, vermiculture, landfills, hazardous waste management

Degradation of xenobiotic in environment: Petroleum hydrocarbons, pesticides, heavy metals.

UNIT- III

(7 Hours)

Water and Air Pollution

Mejor sources of water pollution: Eutropication, acid mine drains, dyeing and tanning.

Techniques to control water pollution: Conventional wastewater treatment- types of sewage, sewage system, safe disposal and its utility.

Natural pathways of degradation: Carbon cycle, Sulphur cycle, Nitrogen cycle and oxygen cycle.

Factors responsible of altering the composition of atmosphere (Deforestation, burning of fossil fuels etc). Techniques to control air pollution.

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DIPLOMA PROGRAM
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UNIT -IV

(7 Hours)

Social Issues and Environmental Conventions

Social Issues and Environment, Environmental conventions Population explosion, Urban problems related to energy, resettlement and rehabilitation of people, their problems and concerns, family welfare programme, human health, women and child welfare.

Environmental ethics & consumerism. Mass tourism and environment. Sustainable tourism.

Stockholm convention, Kyoto Protocol, Montreal Protocol, COP (Conference of Parties), Ramsar Convention, Rio Summit, Convention on Biological Diversity; National Green Tribunal; Relevant case studies, laws and Legislations.

Text/Reference Books:

- 1) Environmental Chemistry: B. K. Sharma
- 2) Industrial Energy Management and Utilization: L.C. Witte, P.S. Schmidt and D.R. Brown (Hemisphere Publishing Corporation, Washington, 1998 8.
- 3) Energy and Environment- NPTEL lecture notes
- 4) Entrepreneurial Development By, S. S. Khanka S. Chand & Co. Ltd. New Delhi, 1999.
- 5) Small- Scale Industries and Entrepreneurship, By, Dr. Vasant Desai, Himalaya Publication.

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B.Tech : IV – SEMESTER

COURSE CODE	COURSE NAME	TH.	TU	PR	CREDITS	EVALUATION		
409BCE24	FULL STACK WEB DEVELOPMENT	-	-	04	02	SEE	CIE	TOTAL
						50	50	100

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] Students will design structured web pages with HTML5, CSS3, and responsive frameworks. [2] To develop interactive and dynamic front-end applications using JavaScript and modern frontend libraries such as React. [3] To impart knowledge on building scalable backend services using Node.js and Express.js, including RESTful API development.	After completing this course, the student will be able to: CO1: Design and implement well-structured, responsive web pages using HTML, CSS, and Bootstrap. CO2: Apply JavaScript, DOM manipulation, and React.js to build interactive frontend components and single-page applications. CO3: Develop backend services using Node.js and Express.js with secure routing, authentication, and middleware integration. CO4: Connect and manage relational and non-relational databases using SQL and MongoDB, and perform CRUD operations. CO5: Build and deploy a real-world full-stack application by integrating frontend, backend, and database, while applying testing and optimization techniques.

Module 1: HTML, CSS & Web Basics

- HTML5 – Structure, Forms, Media, SEO Basics
- CSS3 – Flexbox, Grid, Animations, Responsive Design
- Bootstrap – UI Components & Frameworks

Practical Tasks:

1. Write a program to create a simple webpage using HTML
2. Build a personal portfolio website

Module 2: JavaScript & Frontend Development

- JavaScript (ES6+) – Variables, Loops, Functions, Events
- DOM Manipulation – Event Listeners, Dynamic Content
- AJAX & Fetch API – Handling APIs & JSON

Practical Tasks:

1. Write a program to build a Chat module using HTML CSS and JavaScript?
2. Create a Simple Login form using React JS
3. Develop a weather app using API

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Module 3: Backend Development (Node.js & Express.js)

- **Node.js** – Basics, Modules, NPM
- **Express.js** – Routing, Middleware, Authentication
- **REST API Development** – CRUD Operations
- **JWT & OAuth Authentication**

Practical Tasks:

1. Write a program to create a simple calculator Application using React JS
2. Develop a login system with JWT authentication

Module 4: Database Management

- **SQL (MySQL/PostgreSQL)** – Tables, Queries, Joins
- **NoSQL (MongoDB/Firebase)** – Collections, Schema Design
- **Mongoose ORM** – Schema, Models, Queries

Practical Tasks:

1. Build a user management system
2. Connect a MongoDB database to a web app

Module 5: Full-Stack Project & Best Practices

- **Capstone Project** – Build a real-world full-stack application
- **Testing & Debugging** – Jest, Postman, Chrome DevTools
- **Performance Optimization** – Caching, Lazy Loading

Final Project Idea:

1. **E-commerce website** with user authentication, cart, and payment integration

Text Books:

- 3) **Ultimate Full-Stack Web Development with MERN: Design, Build, Test and Deploy Production-Grade Web Applications with MongoDB, Express, React and NodeJS** Paperback – 1 December 2023, by Nabendu Biswas (Author)
- 4) **Full Stack Web Development: The Comprehensive Guide** (Grayscale Indian Edition) Paperback – 2 August 2023, by Philip Ackermann (Author)

Reference Books:

- 7) **Full Stack Development With Spring Boot 3 And React: Mastering Modern Web Development with Spring Boot 3 and React: Build Scalable, Dynamic Applications from Frontend to Backend** Kindle Edition-28 March 2025 by Simon Tellier (Author) Format: Kindle Edition
- 8) **Full Stack Development with Angular and Spring Boot: Build scalable, responsive, and dynamic enterprise-level web applications** (English Edition) Paperback – 16 August 2024, by Sangeeta Joshi (Author)

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DIPLOMA PROGRAM
CE & ME

B.Tech : IV – SEMESTER

COURSE CODE	COURSE NAME	TH.	TU	PR	CREDITS	EVALUATION		
410BCE24	DATA ANALYTICS WITH PYTHON (MDM-II)	-	-	04	02	SEE	CIE	TOTAL
						50	50	100

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] To understand data analysis concepts and workflows using Python. [2] To use various Python libraries for data manipulation and visualization.	After completing this course, the student will be able to: CO1: Apply techniques for handling missing values, outlier detection, duplicates, and inconsistencies in datasets and Data source. CO2: Demonstrate data preprocessing techniques such as standardization, normalization, encoding, and transformation. CO3: Perform exploratory data analysis (EDA) using summary statistics and data visualization. CO4: Utilize statistical and feature engineering techniques for insightful data analysis.

UNIT -I

(22 Hours)

DATA COLLECTION AND PREPROCESSING

Prerequisites Basics of Python and Libraries , **Introduction to Data Collection** Understand & Apply the Fundamentals of Data Analytics in Real-World Scenarios, **Data Collection** Identify & Evaluate Data Sources, Accessibility, and Permissions for Informed Decision-Making , **Preprocessing** Analyze & Resolve Missing Values, Outlier Detection, and Duplicate Data Challenges.

Task 1: To Study Extracting and Evaluating Data Sources (APIs, Databases, CSVs)

Task 2: To Study Handling Missing Values using Imputation and Dropping Methods

Task 3: To Study Outlier Detection using IQR, and Boxplots

Task 4: To Study Identifying and Removing Duplicate Records from a Dataset

UNIT -II

(16 Hours)

HANDLING CATEGORICAL AND INCONSISTENT DATA

Handling Categorical Implement & Justify Data Standardization, Normalization, and Encoding Techniques , **Inconsistent Data** Transform & Optimize Categorical Data for Better Interpretability and Model Performance. Detect & Address Data Inconsistencies and Handle Class Imbalances Effectively ,

Task 5: To Study Data Standardization and Normalization

Task 6: To Study Data Encoding

Task 7: To Study Handling Categorical Data, Data transformation

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DIPLOMA PROGRAM
CE & ME

UNIT -II

(22 Hours)

EXPLORATORY DATA ANALYSIS AND STATISTICAL INSIGHTS

Design & Construct Design & Construct Feature Engineering Strategies to Enhance Model Performance ,
Interpret & Summarize Data Interpret & Summarize Data Through Descriptive Statistics and Summary Measures, **Examine & Compare Patterns** Examine & Compare Patterns Using Univariate, Bivariate, and Multivariate Analyses,

Task 8: To Study Feature Engineering.

Task 9: To Study Understand the Dataset, Summary Statistics

Task 10: To Study Univariate Analysis

Task 11: To Study Bivariate Analysis

Task 12: To Study Multivariate Analysis

UNIT -II

(08 Hours)

DATA VISUALIZATION AND INTERPRETATION

Visualization Create & Evaluate Line Plots, Bar Charts, Scatter Plots, Box Plots, and Heatmaps.

Interpretation Analyze & Infer Insights from Visual Patterns

Task 13: To Study Data Visualizations - Line plots, bar charts

Task 14: To Study Data Visualizations - scatter plots, box plots, and heat maps

Project (Real-Time Project Development and Assessment of project by Expert)

(06 Hours)

Text Books:

- 1) "Python for Data Analysis" by Wes McKinney
- 2) Written by the creator of Pandas, this book is the best resource for data handling, cleaning, and transformation.
- 3) "Practical Statistics for Data Scientists" by Peter Bruce & Andrew Bruce
- 4) "Data Science for Business" by Foster Provost & Tom Fawcett
- 5) "Data Cleaning: The Ultimate Practical Guide" by David Mertz
- 6) "Hands-On Exploratory Data Analysis with Python" by Suresh Kumar Gorakala & Harish Garg
- 7)

Websites & Online Resources:

Data Extraction, Loading & Handling Missing Values

- 1) Pandas Documentation: <https://pandas.pydata.org/docs/>
- 2) Handling Missing Data with Pandas: <https://towardsdatascience.com/handling-missing-values-in-pandas-dataframe-a16d9f14c603>
- 3) Outlier Detection & Data Cleaning
- 4) IQR Method & Boxplots: <https://towardsdatascience.com/handling-outliers-in-data-analysis-5dceae6bc88b>

Data Standardization, Normalization & Encoding

- 1) Feature Scaling & Normalization: <https://machinelearningmastery.com/normalize-standardize-machine-learning-data-weka/>
- 2) Exploratory Data Analysis (EDA) **EDA using Python:** <https://www.kaggle.com/learn/data-cleaning>
- 3) Univariate, Bivariate, and Multivariate Analysis: <https://towardsdatascience.com/exploratory-data-analysis-8fc1cb20fd15>

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Data Visualization

- 1) Matplotlib Documentation: <https://matplotlib.org/stable/tutorials/>
- 2) Seaborn Guide for Data Visualization: <https://seaborn.pydata.org/>
- 3) Creating Boxplots & Heatmaps: <https://towardsdatascience.com/understanding-boxplots-a331af9f01c>

YouTube Channels for Learning Data Preprocessing & EDA

- 1) StatQuest with Josh Starmer – Explains statistics and data analysis in a fun way.
- 2) Krish Naik – Covers feature engineering, missing values, and outlier detection.
- 3) Data School – Great for Pandas and data preprocessing.
- 4) Corey Schafer – Covers Python, Pandas, and Matplotlib in depth.

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DIPLOMA PROGRAM
CE & ME

B.Tech: IV SEMESTER

COURSE CODE	COURSE NAME	TH	TU	PR	CREDITS	EVALUATION		
411BSH24	COGNITIVE SKILL ENHANCEMENT-LAB	-	-	02	1	ESE	CIE	TOTAL
						25	25	50

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] To develop critical thinking and analytical reasoning for effective problem-solving. [2] To enhance logical reasoning and decision-making skills. [3] To strengthen problem-solving approaches using cognitive strategies and techniques.	After completing this course, the student will be able to: CO1: Understand basic quantitative concepts such as number systems, averages, percentages, and profit and loss for solving fundamental mathematical problems. CO2: Understand critical reasoning and blood relation concepts for basic logical deduction and analysis. CO3: Show accuracy and problem-solving speed through regular practice and competitive activities. CO4: Understand ratio, proportion, mixtures, and alligation concepts to practical problem-solving scenarios.

UNIT I

(12 Hours)

Quantitative Ability (Fundamentals)

Prerequisites: Basic Mathematics, Arithmetic Concepts, Logical Thinking & Problem Solving.

Fundamentals: Basic Calculation Techniques, Problems on Ages, Number System, Averages and Numbers

Task 1: Given 20 arithmetic problems (addition, subtraction, multiplication), the student who solves the most in 5 minutes wins.

Task 2: Solve age-related problems as a group activity.

Task 3: Calculate averages from real-life data. Example: "Find the average score of five students in a recent quiz and discuss the result."

UNIT II

(20 Hours)

Quantitative Ability

Advanced I: Percentages, Profit and Loss, Simple and Compound Interest, Ratio and Proportion, Mixtures and Alligation

Task 4: Multiple-choice quiz on profit and loss.

Task 5: Practical exercise on calculating simple and compound interest.

Task 6: Solve mixture problems in pairs.

Advanced II: Speed Time and Distance, Problems on Trains, Boats and Streams, Time and Word, Pipes and Cistern

Task 7: Teams solve speed, time, and distance problems in a race format.

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Task 8: Visual demonstration of train- related problems.

Example: "Two trains 200 m and 300 m long cross each other in 20 seconds. Calculate their relative speed."

Task 9: Create a scenario with water pipes filling or emptying a tank. Example: "A tank can be filled in 4 hours by Pipe A and emptied in 6 hours by Pipe B. How long will it take to fill the tank if both pipes are open?"

UNIT III

(16 Hours)

Reasoning Ability

Fundamentals: Direction Sense Test, Inequalities, Logical and Number Series, ANS Series, Coding Decoding

Task 10: Use a map to solve direction problems. Example: "You start at Point A, move 5 km north, then 3 km west. Where are you now?"

Task 11: Solve coding-decoding problems to mark correct answers on a bingo card. Example: "If DOG is coded as 4157, how is CAT coded?"

Task 12: Complete number series problems in teams. Example: "What comes next: 2, 4, 8, 16, 32, ___?"

Advanced: Puzzles and Seating Arrangement, Critical Reasoning, Blood Relations, Miscellaneous

Task 13: Hands-on activity using seating charts. Example: "Arrange 8 people in a circular table with specific conditions (A sits to the left of B, C sits opposite D, etc.)."

Task 14: Create and solve blood relation puzzles. Example: "A is the son of B, who is the brother of C. How is A related to C?"

Task 15: Analyze logical statements and debate conclusions.

Example: "Discuss and justify whether the statement 'All cats are animals, but not all animals are cats' is valid."

UNIT IV

(14Hours)

Miscellaneous

Miscellaneous: Miscellaneous topics and questions, Mock Test, Practice Sessions

Task 16: Games based questions: - Use board games like Sudoku, KenKen, and Kakuro to practice logical thinking. Example: "Solve a 9x9 Sudoku puzzle within 10 minutes."

Task 17: Full- length timed test to simulate real exams. Example: A 100-question test covering all units to evaluate overall performance.

Doubt-Solving Session: Open Q&A session for students to ask questions on any topic

Text/Reference Books:

- 1) **Quantitative Aptitude for Competitive Examinations** by R.S. Aggarwal – S. Chand Publications.
- 2) **A Modern Approach to Logical Reasoning** by R.S. Aggarwal – S. Chand Publications
- 3) **Fast Track Objective Arithmetic** by Rajesh Verma – Arihant Publications
- 4) **How to Prepare for Logical Reasoning for CAT** by Arun Sharma – McGraw Hill

Reference Books:

- 1) **Quantitative Aptitude Quantum CAT** by Sarvesh K. Verma – Arihant Publications
- 2) **Analytical Reasoning** by M.K. Pandey – BSC Publications
- 3) **Logical and Analytical Reasoning** by A.K. Gupta – Ramesh Publishing House
- 4) **Puzzles to Puzzle You** by Shakuntala Devi – Orient Paperbacks

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